以下为项目基于文献的事实验证任务代码

本项目文件共包括三个代码文件（ipynb 格式，可使用 jupyter 打开）以及一份 pdf 文件（代码说明）

代码文件包括三部分，分别是：pro, repro-DeBEATa, repro-Deepseek。

pro 文件内为数据预处理代码，运行需加载 bgem3 模型，运行完全部程序可获得处理后的数据文件，分别是：数据内容文件 qasper_docs，数据索引文件：qasper_faiss

```
print("✅ 所有数据已保存到本地：")
print("   1. qasper_faiss.index (向量索引)")
print("   2. qasper_docs.pkl (文本数据库)")
```

Repro-DeBEATa 为 DeBEATa 模型主导分类预测的代码文件。需加载 BGE-M3 模型，bge-rerank-base 模型，BM25 模型，DeBERTa 模型以及前述两份数据文件

由于训练集过程对于两份文件相同，且均由于个人调试均放置于代码后部分（即前置所得为测试集最终数据，如需查看训练信息请运行标题“测试”部分），训练结果仅在此处进行展示。

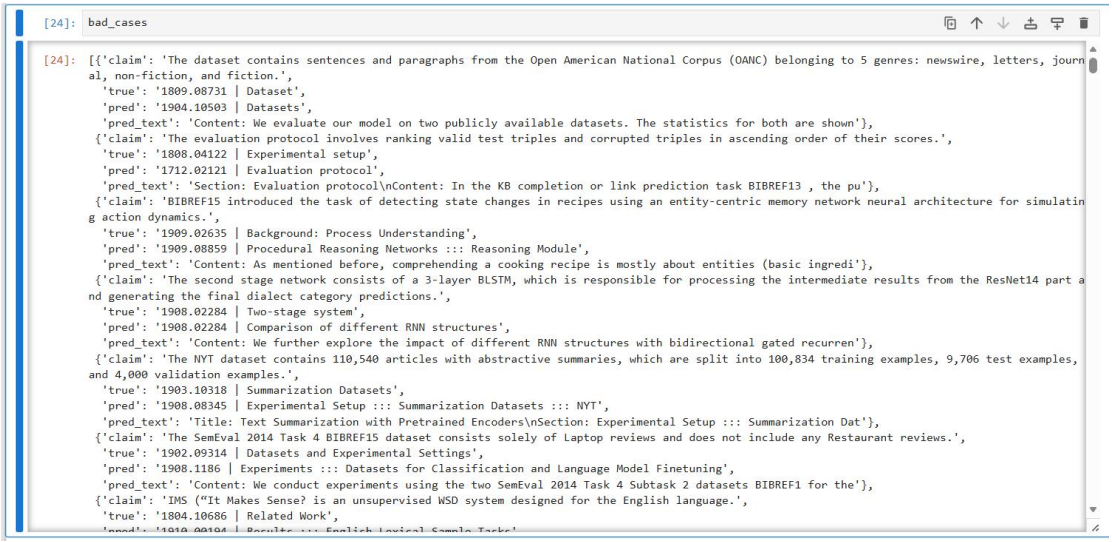
选取 50 条训练数据进行 RAG 预测（DeBERTa 模型）如下：

```
🚀 开始 RAG 全链路严格评估 (Top 3 拼接版)...
--------------------------------------------------
100%|██████████| 50/50 [34:25<00:00, 41.31s/it]

==================================================
📊 【严格模式战报】
样本数量: 50
1. 检索准确率 (Recall@1): 90.00%
2. 端到端严格准确率 (Strict Accuracy): 88.00%
   (要求: 文献ID对 + 章节名对 + 最终判决对)
==================================================

🔍 【丢分原因分析 (前 3 例)】
错题 1:
❌ 推理错误 (Reasoning Error) - 检索已命中，但大模型判错了
   大模型分析: 证据中的第一段明确指出知识图谱通常被解释为由离散的三元组（entityA, relationX, entityB）组成，这与陈述完全一致。此外，没有后续的反转或限制条件否定这一观点。
   预测 Label: 1 | 真实 Label: 0
----------------------------------------
错题 2:
❌ 检索错误 (Retrieval Error)
   预测位置: 1608.01972 | PubMed Experiments
   真实位置: 1608.01972 | Conclusion
----------------------------------------
错题 3:
❌ 检索错误 (Retrieval Error)
   预测位置: 2002.06854 | Experiments and Results ::: Evaluation Metrics and Baselines
   真实位置: 2002.06854 | Experiments and Results ::: Rating Prediction
----------------------------------------
```

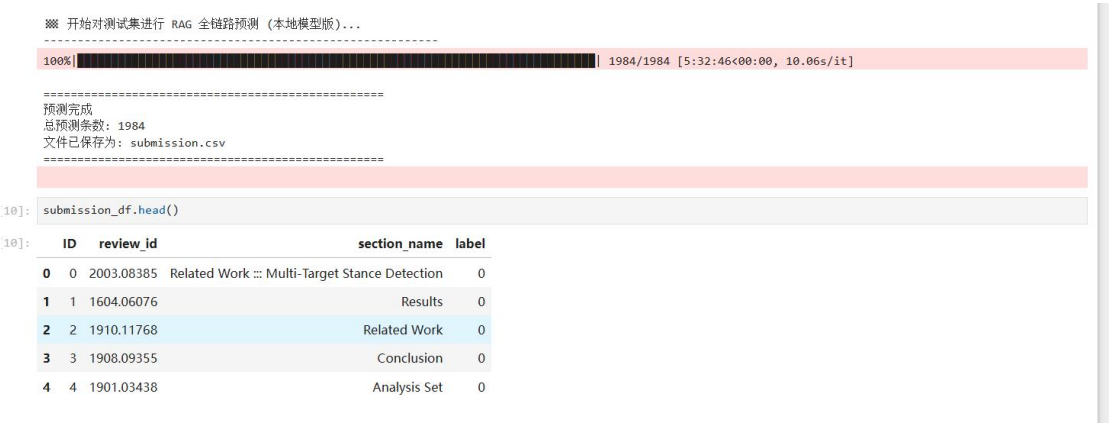
针对 base_case 展示如下：



```
[24]: bad_cases

[24]: [{'claim': 'The dataset contains sentences and paragraphs from the Open American National Corpus (OANC) belonging to 5 genres: newswire, letters, journ
      al, non-fiction, and fiction.',
        'true': '1809.08731 | Dataset',
        'pred': '1904.10503 | Datasets',
        'pred_text': 'Content: We evaluate our model on two publicly available datasets. The statistics for both are shown'},
       {'claim': 'The evaluation protocol involves ranking valid test triples and corrupted triples in ascending order of their scores.',
        'true': '1808.04122 | Experimental setup',
        'pred': '1712.02121 | Evaluation protocol',
        'pred_text': 'Section: Evaluation protocol\nContent: In the KB completion or link prediction task BIBREF13 , the pu'},
       {'claim': 'BIBREF15 introduced the task of detecting state changes in recipes using an entity-centric memory network neural architecture for simulatin
      g action dynamics.',
        'true': '1909.02635 | Background: Process Understanding',
        'pred': '1909.08859 | Procedural Reasoning Networks ::: Reasoning Module',
        'pred_text': 'Content: As mentioned before, comprehending a cooking recipe is mostly about entities (basic ingredi'},
       {'claim': 'The second stage network consists of a 3-layer BLSTM, which is responsible for processing the intermediate results from the ResNet14 part a
      nd generating the final dialect category predictions.',
        'true': '1908.02284 | Two-stage system',
        'pred': '1908.02284 | Comparison of different RNN structures',
        'pred_text': 'Content: We further explore the impact of different RNN structures with bidirectional gated recurren'},
       {'claim': 'The NYT dataset contains 110,540 articles with abstractive summaries, which are split into 100,834 training examples, 9,706 test examples,
      and 4,000 validation examples.',
        'true': '1903.10318 | Summarization Datasets',
        'pred': '1908.08345 | Experimental Setup ::: Summarization Datasets ::: NYT',
        'pred_text': 'Title: Text Summarization with Pretrained Encoders\nSection: Experimental Setup ::: Summarization Dat'},
       {'claim': 'The SemEval 2014 Task 4 BIBREF15 dataset consists solely of Laptop reviews and does not include any Restaurant reviews.',
        'true': '1902.09314 | Datasets and Experimental Settings',
        'pred': '1908.1186 | Experiments ::: Datasets for Classification and Language Model Finetuning',
        'pred_text': 'Content: We conduct experiments using the two SemEval 2014 Task 4 Subtask 2 datasets BIBREF1 for the'},
       {'claim': 'IMS (“It Makes Sense? is an unsupervised WSD system designed for the English language.',
        'true': '1804.10686 | Related Work',
        'pred': '1910.00194 | Results ::: English Lexical Sample Tasks'
```

使用 DeBERTa 模型所得结果如下：



```
🚀 开始对测试集进行 RAG 全链路预测（本地模型版）...
------------------------------------------------------------
100%|██████████| 1984/1984 [5:32:46<00:00, 10.06s/it]

==================================================
预测完成
总预测条数：1984
文件已保存为：submission.csv
==================================================
```

```
[10]: submission_df.head()
```

[10]:

	ID	review_id	section_name	label
0	0	2003.08385	Related Work ::: Multi-Target Stance Detection	0
1	1	1604.06076	Results	0
2	2	1910.11768	Related Work	0
3	3	1908.09355	Conclusion	0
4	4	1901.03438	Analysis Set	0

repro-Deepseek 文件内为使用 Deepseek 模型代码，其中 api 与 url 未列出，如需运行请自行填写相关大模型信息（注意修改大模型数据）：

```
# 接入模型(由于个人隐私原因不列出相关API URL)
API_KEY =
BASE_URL =
MODEL_NAME = "deepseek-ai/DeepSeek-V3"
```

整体测试结果运行时长在 12-16 个小时，如有需要请自行把握时间运行。